COMP8620

Advanced Topics in AI: Intelligent Robotics

Assignment 2 Report

Semester 2, 2026

Student	Rui Luo
UID	u8076655

The Australian National University

Generative AI Disclosure

Two generative-AI tools were used. OpenAI Codex was used to prepare the report template, review and revise Parts A and B, implement the initial Part B solver and tests, design and run experiments, and prepare pseudocode and figures. Claude (Anthropic, accessed via Claude Code) was used to review and revise the solver, redesign and rerun experiments, analyse the results, and revise the report, tables and figures. All prompts are reproduced verbatim in their original language in `AI_prompts.txt`, included in the submitted source archive.

Part A: Theory (30 points)

Question 1: Two-state MDP abstraction (10 points)

No.

Let $m = \lceil k/2 \rceil$, and define

$$\phi(s) = \begin{cases} L, & 1 \le s \le m, \\ R, & m+1 \le s \le k. \end{cases}$$

An exact MDP abstraction requires a transition function depending only on the abstract state and action while preserving the original motion probabilities. In particular, for every action a, transition consistency requires

$$\phi(s_1) = \phi(s_2) \implies \sum_{u:\, \phi(u)=R} T(s_1, a, u) = \sum_{u:\, \phi(u)=R} T(s_2, a, u).$$

For $k > 4$, consider $s_1 = m - 1$ and $s_2 = m$, both in L. Under $a = \text{moveRight}$,

$$m-1 \xrightarrow{0.7} m \in L, \qquad m-1 \xrightarrow{0.3} m-2 \in L,$$
$$m \xrightarrow{0.7} m+1 \in R, \qquad m \xrightarrow{0.3} m-1 \in L.$$

Hence,

$$P(R \mid s_1, a) = \sum_{u=m+1}^{k} T(m-1, a, u) = 0,$$
$$P(R \mid s_2, a) = \sum_{u=m+1}^{k} T(m, a, u) = T(m, a, m+1) = 0.7.$$

Since $0 \neq 0.7$, transition consistency fails: no single $T'(L, \text{moveRight}, R)$ can represent both states exactly.

Therefore, the proposed abstraction cannot preserve the original motion uncertainty and is not an exact MDP abstraction.

Question 2: Alpha-vector dominance (10 points)

Let $p = b(s_1)$. Since $b(s_1) + b(s_2) = 1$,

$$b = (p, 1-p), \qquad 0 \le p \le 1.$$

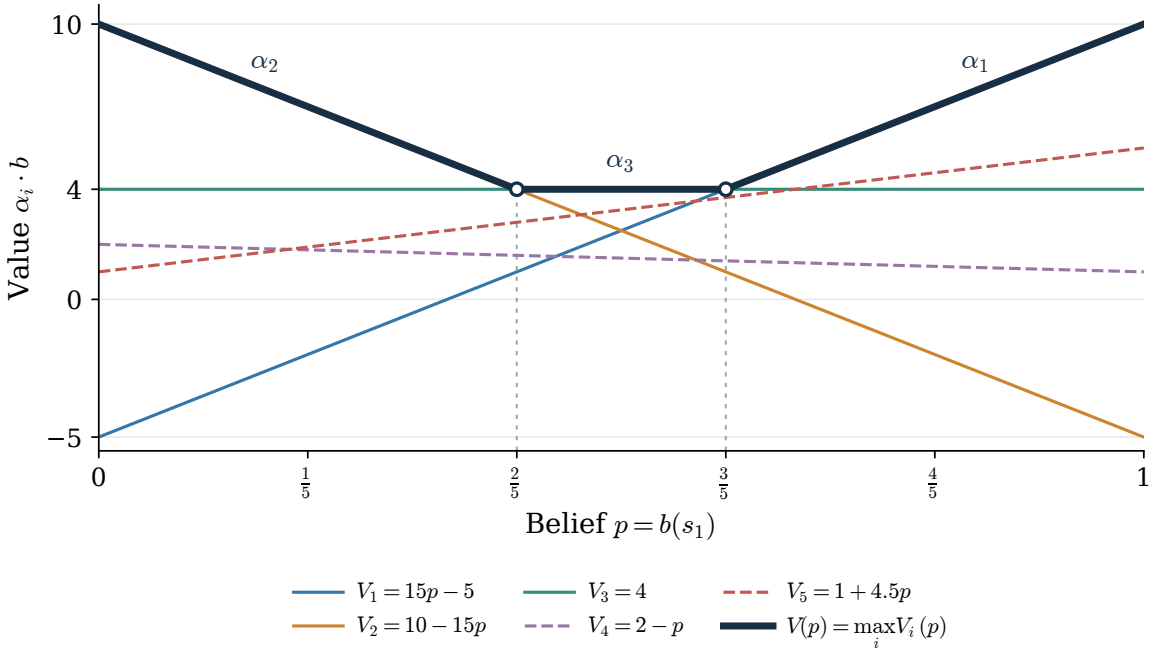
For each $\alpha_i \in \Gamma$, let $V_i(p) = \alpha_i \cdot b$. Then

$$\begin{aligned} V_1(p) &= 10p - 5(1-p) = 15p - 5, \\ V_2(p) &= -5p + 10(1-p) = 10 - 15p, \\ V_3(p) &= 4p + 4(1-p) = 4, \\ V_4(p) &= p + 2(1-p) = 2 - p, \\ V_5(p) &= 5.5p + (1-p) = 1 + 4.5p. \end{aligned}$$

(a) Maximal belief intervals (4 points)

The upper envelope is

$$V(p) = \max_{\alpha_i \in \Gamma} V_i(p).$$



The relevant intersections are

$$\begin{aligned} V_2(p) = V_3(p) &\iff 10 - 15p = 4 \iff p = \frac{2}{5}, \\ V_1(p) = V_3(p) &\iff 15p - 5 = 4 \iff p = \frac{3}{5}. \end{aligned}$$

The remaining vectors are never maximal, since

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad 0 \le p \le 1,$$

and, using $V_1(p) - V_5(p) = 10.5p - 6$,

$$\begin{cases} V_5(p) \le 1 + 4.5 \times \frac{3}{5} = 3.7 < 4 = V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_1(p) - V_5(p) \ge 10.5 \times \frac{3}{5} - 6 = 0.3 > 0, & \frac{3}{5} \le p \le 1. \end{cases}$$

Since V_2 decreases, V_1 increases, and $V_3 = 4$,

$$V(p) = \begin{cases} 10 - 15p, & 0 \le p < \frac{2}{5}, \\ 4, & \frac{2}{5} \le p \le \frac{3}{5}, \\ 15p - 5, & \frac{3}{5} < p \le 1. \end{cases}$$

Therefore, the maximal belief intervals are

$$\boxed{\begin{aligned} \alpha_1 &: \left[\frac{3}{5}, 1\right], \\ \alpha_2 &: \left[0, \frac{2}{5}\right], \\ \alpha_3 &: \left[\frac{2}{5}, \frac{3}{5}\right], \\ \alpha_4, \alpha_5 &: \varnothing. \end{aligned}}$$

At $p = \frac{2}{5}$, α_2 and α_3 tie; at $p = \frac{3}{5}$, α_1 and α_3 tie.

(b) Pointwise dominance (6 points)

An α-vector can be removed if, at every belief, another vector has strictly greater value:

$$\forall b \in B, \quad \exists \alpha' \in \Gamma : \alpha \cdot b < \alpha' \cdot b.$$

For α_4,

$$V_4(p) = 2 - p < 4 = V_3(p), \qquad \forall p \in [0, 1].$$

Thus, α_4 is strictly dominated by α_3.

For α_5, the dominating vector may depend on p. From part (a),

$$\begin{cases} V_5(p) < V_3(p), & 0 \le p \le \frac{3}{5}, \\ V_5(p) < V_1(p), & \frac{3}{5} < p \le 1. \end{cases}$$

These intervals cover $[0, 1]$, so α_5 also satisfies the removal condition.

The remaining vectors α_2, α_3, and α_1 are uniquely maximal at $p = 0$, $p = \frac{1}{2}$, and $p = 1$, respectively. Hence, none satisfies the removal condition.

Therefore,

$$\boxed{\Gamma_{\text{remove}} = \{\alpha_4, \alpha_5\}, \qquad \Gamma_{\text{keep}} = \{\alpha_1, \alpha_2, \alpha_3\}.}$$

Question 3: MDP versus POMDP for pedestrian avoidance (10 points)

Let c_t, v_t be the car's position and speed, and p_t^i the position of pedestrian i. Define

$$x_t = (c_t, v_t, p_t^1, \ldots, p_t^n).$$

Positions are observed accurately, and the car measures its own speed. It chooses an action a_t (accelerate, maintain speed, or brake) to reach G while avoiding collisions.

(a) Scenario suitable for an MDP

Suppose pedestrians follow known stochastic movement rules: the probabilities of crossing or continuing along the sidewalk depend only on the current observable configuration, with no persistent hidden intention. These memoryless dynamics give

$$P(x_{t+1} \mid x_{0:t}, a_{0:t}) = P(x_{t+1} \mid x_t, a_t).$$

Thus, x_t is Markov and fully observed, so an MDP with state $s_t = x_t$ is suitable despite the stochastic pedestrian motion.

(b) Scenario suitable for a POMDP

Now suppose each pedestrian has a persistent but unobservable destination g_i, such as crossing the road or continuing along the sidewalk. Let $\mathbf{g} = (g_1, \ldots, g_n)$, and assume motion is Markov conditional on these destinations. The complete state and observation are

$$s_t = (x_t, \mathbf{g}), \qquad o_t = x_t.$$

At the same observed configuration, two possible joint destinations can yield different motion distributions:

$$P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(1)}) \neq P(x_{t+1} \mid x_t, a_t, \mathbf{g}^{(2)}).$$

Different motion histories can produce the same x_t but different beliefs about $\mathbf{g}$. Hence, x_t alone need not be Markov. The car maintains

$$b_t(\mathbf{g}) = P(\mathbf{g} \mid x_{0:t}, a_{0:t-1})$$

and predicts motion using

$$P(x_{t+1} \mid x_t, b_t, a_t) = \sum_{\mathbf{g}} P(x_{t+1} \mid x_t, a_t, \mathbf{g})\, b_t(\mathbf{g}).$$

The belief is updated as new movements are observed. A POMDP is suitable because it uses both the observed configuration and the belief over hidden destinations to select driving actions.

Part B: Multi-drone planning under uncertainty (35 points)

Question 1: MDP formulation (3 points)

Let $\mathcal{G} = \{0, \dots, n_x - 1\} \times \{0, \dots, n_y - 1\} \times \{0, \dots, n_z - 1\}$, with obstacles O and known drone goals g_i. The finite-horizon MDP is $(\mathcal{S}, \mathcal{A}, T, R, \gamma)$.

States and actions. A live state is

$$s = ((p_i, r_i)_{i=1}^{N}, t), \qquad 0 \le t < H,$$

where p_i is drone i's cell and r_i indicates that it has reached its goal. Reached drones remain at their goals and occupy those cells. The state space also contains an absorbing state $\perp$, entered after a collision, all goals being reached, or the step limit H. Each active drone commands one of $B = 26$ neighbour offsets (or $B = 8$ with fixed altitude), giving $\mathcal{A} = \{0, \dots, B-1\}^N$. There is no hover action; reached drones ignore their commands.

Transitions. Let $q_i(y_i \mid p_i, r_i, a_i)$ be the supplied, normalised single-drone target law (Appendix A.1). For a reached drone, $q_i(y_i \mid p_i, 1, a_i) = \mathbf{1}\{y_i = p_i\}$. Raw targets $y = (y_1, \dots, y_N)$ are conditionally independent. Let $F(s, y)$ update positions and flags, increment t, and return $\perp$ on termination. Then

$$T(s, a, s') = \sum_{y : F(s,y) = s'} \prod_{i=1}^{N} q_i(y_i \mid p_i, r_i, a_i), \qquad T(\perp, a, \perp) = 1.$$

An obstacle hit or shared returned cell terminates the mission; cell swaps and diagonal crossings are permitted by the supplied collision rule.

Reward and objective. For a sampled outcome y, let K_O count obstacle hits, K_V count drones in shared-cell groups, and Δr_i indicate a newly reached goal. The simulator returns

$$\widetilde{r}(s, a, y) = N c_{\text{step}} + C(K_O + K_V) + G \sum_i \Delta r_i, \qquad c_{\text{step}} < 0,\ C < 0,\ G > 0.$$

Thus $R(s, a) = \mathbb{E}_y[\widetilde{r}(s, a, y) \mid s, a]$ and $R(\perp, a) = 0$. The terminal transition receives its reward before absorption. The objective is

$$\max_{\mu}\ \mathbb{E}_{\mu}\left[\sum_{t=0}^{\tau-1} \gamma^t \widetilde{r}_t\right], \qquad \tau = \min\{T_{\text{success}}, T_{\text{collision}}, H\}.$$

This encourages fast completion and penalises collisions.

Question 2: Approximate online solver (4 points)

I use UCT/MCTS with progressive widening, re-planned after every executed step. The supplied generative model samples outcomes without enumerating the state space or all $26^4 = 456{,}976$ joint actions.

Guidance: single-drone value iteration. Before planning, each drone's MDP is solved alone, ignoring the other drones, by finite-horizon value iteration on the supplied slip law (Week 5 MDP-solving lecture, slide 9). With per-drone reward $r_i = c_{\text{step}} + C\,\mathbf{1}\{\text{crash}\} + G\,\mathbf{1}\{\text{arrival}\}$,

$$Q_{i,h}(p, a) = \mathbb{E}[r_i + \gamma V_{i,h-1}(p') \mid p, a], \qquad V_{i,h}(p) = \max_a Q_{i,h}(p, a), \qquad V_{i,0} = 0,$$

for $h = 1, \dots, H$, where crash and arrival end that drone's problem. $Q_{i,h}$ ranks commands and $V_{i,h}$ evaluates rollout cutoffs, both at the number h of steps actually left. Solving each drone

alone is our approximation for the joint MCTS, not a solution of the joint problem. For a command from p_i towards q (action a_q) with h steps left, the score, in step-cost units and lower being better, is

$$-\frac{Q_{i,h}(p_i, a_q)}{\eta} + 0.025\|q - g_i\|_1 + \lambda \ell(p_i) \sum_{\substack{k \neq i \\ r_k = 0}} \sigma(p_k) \mathbf{1}\{q \in \mathcal{N}(p_k)\}.$$

Here $\eta = \max\{|c_{\text{step}}|, 10^{-3}(G + |C|), 10^{-9}\}$ is the bounded step-cost unit; ℓ and σ are the normalised intended and per-neighbour slip probabilities, and λ converts collision loss into step-cost units (Appendix A.1). The last term is an inter-drone risk surrogate. $\mathcal{N}$ contains physical neighbours at the allowed altitude. Intended-target filtering removes commands whose intended cell is outside the grid or an obstacle (the simulator would accept them), and frozen or already reserved targets incur an additional 10^9. GREEDY processes drones in index order and minimises this score; SAMPLE is a truncated softmax proposal: random drone order, sampling among the four cheapest commands within 1.05 of the minimum with weights $e^{-3(\text{score}-\min)}$. EXPLORE samples uniformly from free, in-grid, unreserved targets when available. An earlier guidance (A) replaced $-Q_{i,h}/\eta$ by risk-weighted Dijkstra target costs plus a clearance tie-breaker $0.3/\max\{1, d(q)\}$; development round 3 adopted value guidance, and Appendix A.8 compares both on S1, S2, S4, S5 and the additional validation suites.

Figure 1 separates distance, slip risk and value. In the cluttered map, the value-guided route leaves the high-risk block that the step distance would cut through. Behind the wall, every cell carries the unavoidable cost of crossing the one gap, and the saved run heads for the gap. These examples explain the guidance; they do not establish a performance improvement.

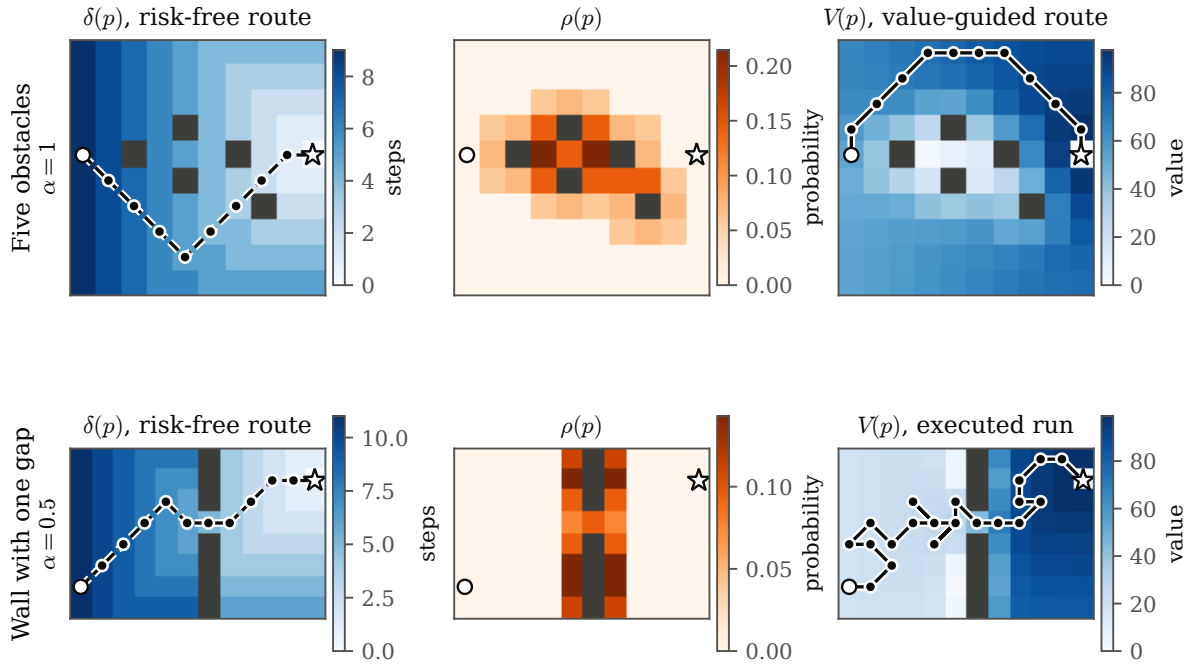


Figure 1: Guidance maps for two fixed-altitude single-drone cases. Left: shortest-step distance $\delta(p)$ and the risk-free greedy route. Middle: one-step obstacle-slip probability $\rho(p)$ for a free target command. Right: single-drone value $V_{i,H}(p)$ with the value-guided greedy route (top) or a saved seed-0 execution of the submitted solver (bottom). Dark cells: obstacles; circle: start; star: goal. Greedy routes assume no slips; colour scales differ between panels.

The four MCTS components

(1) **Selection.** Try unvisited admitted actions first; otherwise maximise

$$Q(s,a) + c_{\text{UCT}}\sqrt{\frac{\log(N(s)+1)}{N(s,a)}}, \qquad c_{\text{UCT}} = 0.35.$$

Returns are normalised by $\kappa = \max\{1, N(G + 2|C|)\}$.

(2) **Expansion.** The action limit is $\min\{\lceil 0.5\sqrt{N(s)+1}\rceil, \prod_{i:r_i=0} |U_i^{\text{adm}}|\}$. Here U_i^{adm} contains commands towards free, in-grid neighbours. Admit GREEDY first, then EXPLORE with probability 0.15 and the truncated softmax SAMPLE otherwise (at most 12 attempts for a new action). Each distinct sampled successor has its own child.

(3) **Simulation.** A greedy rollout with ε exploration runs for at most eight steps, choosing EXPLORE with probability 0.1, otherwise GREEDY. Terminal outcomes have zero continuation value; other cutoffs with h steps left add $\widehat{V}(s,h)/\kappa$, where the unnormalised heuristic leaf value is

$$\widehat{V}(s,h) = \sum_{i:r_i=0} V_{i,h}(p_i) + c_{\text{step}}\,|\{i : r_i = 1\}| \sum_{j=0}^{D-1} \gamma^j, \qquad D = \min\{h, \max_{i:r_i=0} \delta_i\},$$

where the second term charges drones already parked at their goals the step cost while others fly (δ_i: breadth-first distance). A drone that frozen goal occupants disconnect from its goal contributes $c_{\text{step}} \sum_{j<h} \gamma^j$ instead (checked on maps of at most 20,000 cells). Tree depth is capped at eight.

(4) **Backpropagation.** A sample-mean return backup with heuristic leaf evaluation: back up $v = R_{\text{sample}}/\kappa + \gamma v'$; increment edge and node visits and update the edge mean and sample variance using Welford's recurrence.

Root decision. Let a_g be the greedy action and $\hat{\sigma}^2$ the pooled within-action variance. Among visited actions, define

$$m(a) = Q(a) - Q(a_g) - z_{\text{sw}}\hat{\sigma}\sqrt{\frac{1}{N(a)} + \frac{1}{N(a_g)}}, \qquad z_{\text{sw}} = 1.$$

Execute the action with largest positive $m(a)$; otherwise keep a_g. With no completed trial, use a_g directly. This is heuristic switching based on an estimated standard-error margin; it has no formal policy-improvement guarantee. Parameters, the guidance and this root rule were selected on 16 separate development layouts; executing the largest-mean-Q action instead raised L there (Appendix A.2).

Listing 1: Online UCT; rollout and admission rules are defined above.

```
PROCEDURE PLAN(s, budget, K = 2000)
    deadline <- NOW + max(0, 0.97 * min(budget, 10) - 0.002)
    h <- H - t; a_g <- GREEDY(s, h); root <- empty node; sims <- 0
    WHILE NOW < deadline AND sims < K AND h > 0 AND not all reached:
        TRY: SEARCH(root, s, h, 0); sims += 1
        ON Deadline: BREAK
    RETURN root decision above, or a_g if no trial completed

PROCEDURE SEARCH(node, s, h, depth)
    IF h = 0 OR all reached: RETURN 0
    IF depth >= 8: RETURN ROLLOUT(s, h)
    admit an action under the expansion rule, if permitted
    a <- unvisited action first, otherwise argmax UCB
    (s', r, done) <- SIMULATE(s, a) // check deadline before call
    IF done: v <- 0
    ELSE IF child(node, a, s') is new:
        create child; v <- ROLLOUT(s', h - 1)
    ELSE: v <- SEARCH(child(node, a, s'), s', h - 1, depth + 1)
    v <- r / kappa + gamma * v
    update visits, mean and variance; RETURN v
```

ROLLOUT discounts sampled rewards and the terminal/cutoff value defined above. It may stop at the deadline and return a leaf estimate; a deadline exception before a simulator call aborts that trial. The planner tracks the remaining horizon because `simulate` does not enforce the execution step limit.

Question 3: Implementation and demonstration (8 points)

`run_u8076655.py` implements B2 using the supplied MultiDroneUnc simulator's unchanged dynamics, with independent planning and execution RNGs. The planner uses `get_config()`, `simulate()` and `num_actions`. AST checks compare the core simulator methods against supplied revision `ea3cbfe` (`UPSTREAM.json`); only plotting imports/hooks were adjusted for headless execution. VI preprocessing is recorded separately as `setup_seconds`, outside the per-step planning times. From the extracted archive's top-level folder:

Reproduction commands.

```
python -m pip install -r requirements.txt
python run_u8076655.py --config <scenario.yaml>

# Optional flags; full descriptions are in README.md.
python run_u8076655.py --config <scenario.yaml> \
  [--planning-time SECONDS] [--max-simulations K] [--seed SEED] \
  [--visualize] [--quiet] [--output TRACE.json]

# Example with the optional Vedo viewer.
python run_u8076655.py --config environments/demo_four_3d.yaml --visualize
```

Each executed step prints one line to standard error; a JSON summary goes to standard output. Excerpt from the open 2D validation run below:

```
step   1/80 | 0/1 at goal | reward    -1.0 | plan  7.42s  2000 sims | (1,3,0)
step   2/80 | 0/1 at goal | reward    -1.0 | plan  7.26s  2000 sims | (2,3,0)
...
step  14/80 | 1/1 at goal | reward    99.0 | plan  1.84s  2000 sims | (11,5,0)*
success: True (success), steps: 14, total discounted reward: 64.58
```

Planning is serial, with a 9.5 s allowance and 2,000-trial cap; the deadline is checked before each simulator call. All 45 unit tests pass. Local single-process runs with the default 9.5-second allowance (seed 0) give:

Task	Status	Steps	Sims/step	Mean (s)	Max (s)
Open 2D, 1 drone, 8 actions	success	14	2,000	4.53	7.42
Supplied example, 2 drones	success	17	2,000	5.59	7.48
Strong wind, 1 drone, $\alpha = 3$	success	30	2,000	4.72	6.34
Four 3D, 5 obstacles	success	13	1,986	6.27	9.21

Question 4: Experimental investigation (20 points)

Experimental design

The main evaluation uses 2,000 simulations per decision; S3 includes ablations and a zero-search policy, and S4 varies the simulation cap. Experimental runs use a relaxed wall-clock cap to isolate search effort; the demo stops at the first of its time and trial limits. In the four local single-process B3 runs the 2,000-trial cap ended 73 of 74 decisions, and the one clock-bound decision still completed 1,815 simulations (largest decision 9.21 s). Slower hardware could end decisions on the clock with fewer simulations; S4 finds no consistent completion improvement from 16 to 2,000 simulations on four tasks. Its intervals do not establish equivalence or robustness to a binding deadline on other hardware. Wall times logged during the parallel experiments reflect CPU sharing between worker processes, not demo decision times. Conditions share execution seeds 0–19 unless noted, with comparisons paired within each scenario. Development layouts were excluded from evaluation. Only independent experimental episodes ran in parallel; planning within each episode was serial.

Set	Controlled scenarios	Question and expectation
S1 (540)	3^3 factorial: $12 \times n_y \times 3$ grids, $n_y = 5, 10, 20$; 6/18/26 commands per drone ($\|A\| = 36/324/676$); 0/2/5 nested obstacles. Two drones, $\alpha = 0.25$, $H = 60$.	Rank the three factors and test interactions. Off-route cells should have little effect; richer control may help; clutter adds detours and slip risk.
S2 (160)	S1 clutter layouts at $\alpha = 0.01$ with 6/26 commands; duplicate command labels $\times 2, \times 4$ without changing physics.	Separate geometric detours from noise, and control authority from branching.
S3 (1,040)	Six ablations (including two root rules) and the search-free policy at all eight low/high S1 corners. Uniform proposals use ten seeds; others use twenty.	Measure how component effects change with each complexity factor.
S4 (400)	16/64/256/1,024/2,000 simulations on open 1-drone, strong-wind 1-drone, cluttered 2-drone and 3D 4-drone tasks.	Test whether additional search improves completion as tasks get harder.
S5 (160)	$N = 1, 2, 3, 4$ crossed with $\alpha = 0.05, 0.25, 1, 2$; one random $8 \times 6 \times 3$ layout per pair, ten seeds each.	Check performance beyond the factorial and joint growth of state/action spaces.

Table 1: Evaluation sets; parentheses give episode counts.

A further 360 regression, 520 search-free baseline and 280 supplementary runs (140 full solver, 140 search-free) provide additional validation (Appendix A.9); the complete evaluation contains 3,460 episodes.

Metrics. With F free cells, there are up to F^N positional configurations and B^N joint commands. Success means all goals reached within H without collision. Penalised completion length is $L =$ steps on success and $L = H$ on failure. Success proportions have Wilson 95% intervals; continuous means have Student-t intervals $\bar{x} \pm t_{0.975,n-1} s/\sqrt{n}$. High-minus-low factor effects average the other factors within each seed; component effects pair each ablation with the full solver before averaging corners. Their 95% intervals use seed blocks. Cross-suite guidance and baseline differences instead use percentile bootstrap intervals from 20,000 resamples of whole execution-seed clusters, preserving dependence across conditions; repeated conditions are counted once in the guidance aggregate. Supporting means and intervals are in Appendix A.

Execution environment. Episodes ran on macOS-26.6.2 (`x86_64`), Python 3.12.9, NumPy 1.26.4, SciPy 1.15.1, PyYAML 6.0.2. Wall times depend on hardware and load; a fixed simulation cap controls search effort only when the relaxed experimental deadline does not bind.

S1: effects of the three complexity factors

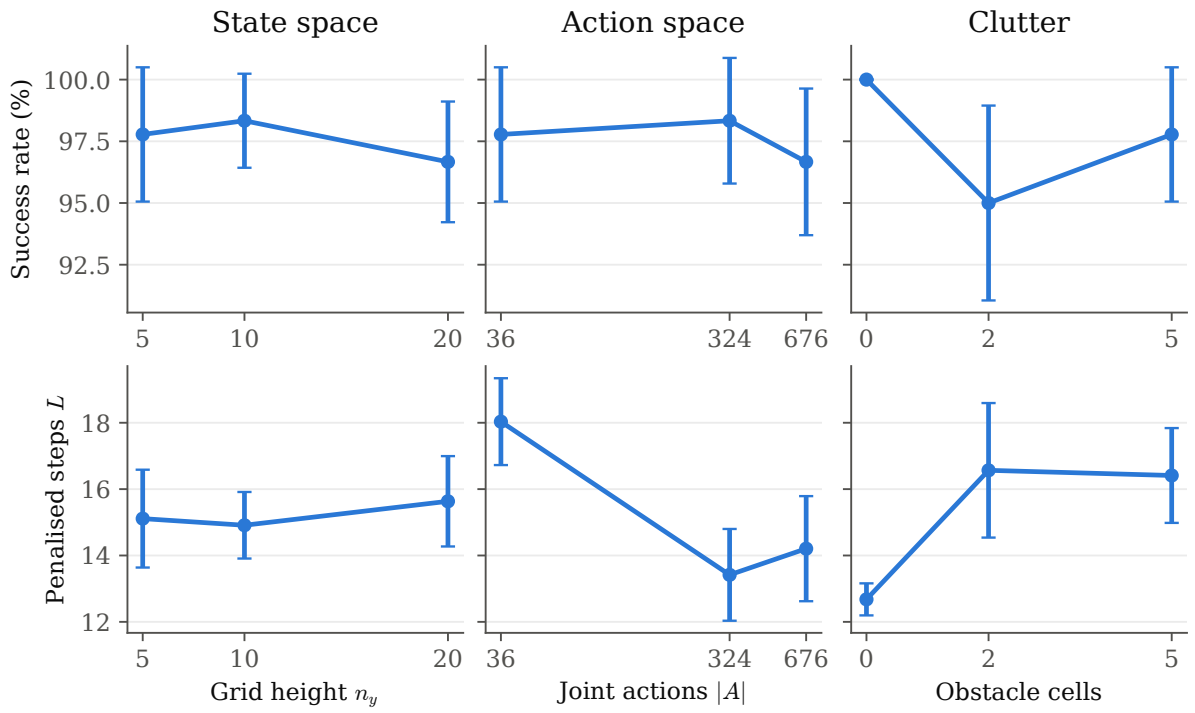


Figure 2: S1 marginal means for success and penalised completion length L, averaging the other factors within each seed. Bars show 95% intervals over 20 seed blocks. Action-space sizes count joint commands.

Factor (high minus low)	ΔL (95% CI)	Δsuccess (pp; 95% CI)
State space (n_y 5→20)	+0.5 [-0.9, +1.9]	-1.1 [-4.0, +1.8]
Action space ($\|A\|$ 36→676)	-3.8 [-5.8, -1.9]	-1.1 [-5.2, +3.0]
Clutter ($\|O\|$ 0→5)	+3.7 [+2.5, +4.9]	-2.2 [-4.9, +0.5]

Table 2: Paired factor effects; positive ΔL indicates degradation.

Clutter causes the largest degradation within these tested levels. Its success effect is -2.2 pp $[-4.9, +0.5]$; no factor's success interval excludes zero, so the ranking rests on L (clutter minus state $+3.2$ $[+1.7, +4.7]$, clutter minus action $+7.6$ $[+5.3, +9.8]$). The solver succeeds in 97.6% of S1 episodes; all 13 failures are collisions and none reaches the horizon.

State-space padding has little observed effect. For $n_y = 10$ versus 20, 102 of 180 pairs match in steps, return and outcome. This tests off-route padding, not longer routes or additional drones.

Richer commands shorten routes. Changing 6 to 26 directions reduces L; the success difference is unresolved. The action–clutter interaction is -3.6 $[-4.5, -2.7]$ steps, consistent with diagonal/vertical moves easing obstacle avoidance. The state–action interaction is also resolved (-5.0 $[-8.3, -1.7]$): richer commands help more in the wide $n_y = 20$ grid, plausibly because the

boundary of the narrow grid truncates lateral slips that diagonal commands would otherwise correct. The state–clutter interaction is not resolved ($[-2.0, +3.5]$).

S2: geometry, noise and action branching

At $n_y = 10$, the clutter effect averaged over the two command sets is +4.5 steps $[+2.1, +6.8]$ at $\alpha = 0.25$, versus +1.0 $[+0.7, +1.3]$ at $\alpha = 0.01$, where one of 120 episodes fails. Thus geometry still adds detour cost, while noise substantially increases the observed clutter penalty. The supplied transition law raises slip probability near obstacles.

Duplicating command labels up to $|A| = 10{,}816$ leaves success at 19/20 and mean L within 0.3 steps. The paired interval $[-7.5, +6.9]$ includes zero; failures occur on different seeds and contribute to its width. The admitted root-action count stays near 23. This separates pure label branching from the extra control authority in S1. Detailed controls, including the visit share of the *executed* action, appear in Appendix A.4.

S3: component effects as complexity increases

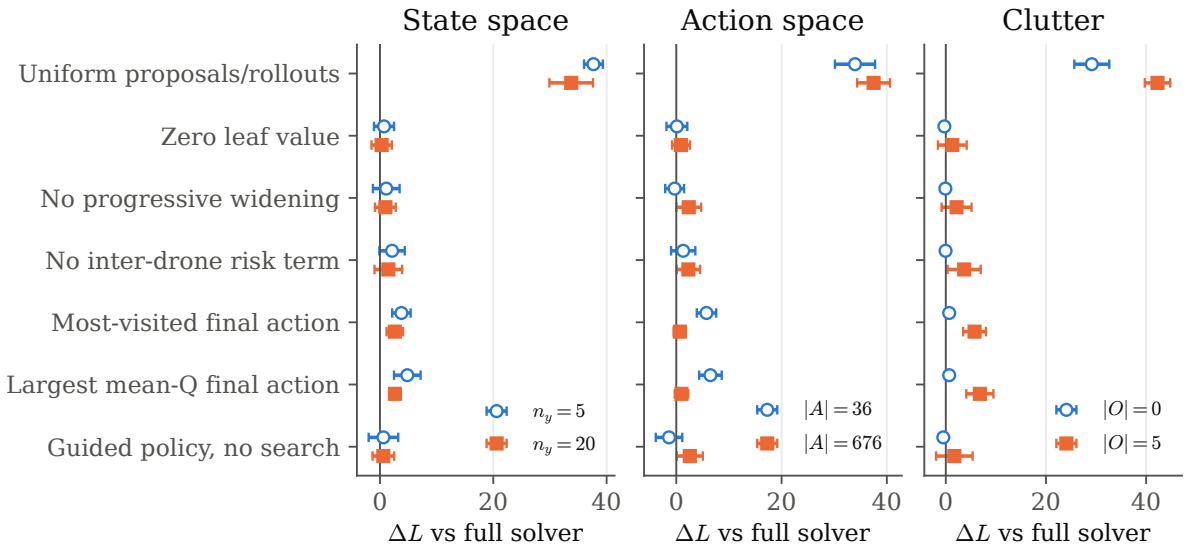


Figure 3: Paired change in L after removing a component at low/high factor levels (open/filled markers); 95% seed-block intervals. Positive values favour the full solver.

Guidance in expansion and rollout has the largest measured effect. Uniform proposals/rollouts raise L by +35.7 $[+33.6, +37.8]$ and change success by −50.0 pp. The loss grows from +29.1 without obstacles to +42.3 with five; the clutter interaction excludes zero ($[+8.7, +17.5]$), whereas its state/action interactions do not. This ablation removes both route guidance and admissible-target filtering.

Root choice also matters: most-visited selection raises L by +3.2 $[+2.1, +4.3]$, and executing the largest-mean-Q action raises it by +3.7 $[+2.4, +5.1]$. Both increases are larger with clutter (interaction intervals $[+2.8, +7.4]$ and $[+3.3, +8.9]$) and smaller with richer commands ($[-7.1, -3.0]$ and $[-7.8, -3.2]$). These comparisons support the chosen switching rule on these tasks; they do not make its estimated margin a calibrated confidence bound.

Widening and the leaf value have no statistically resolved effects across these corners (+1.0 $[-0.5, +2.5]$ and +0.5 $[-1.0, +2.0]$; full estimates in Appendix A.5). Routes are roughly eleven steps, comparable to the sampled tree depth plus eight-step rollout, which may limit the leaf value's contribution. **The inter-drone risk term** does matter: removing it raises L by +1.8 $[+0.1, +3.5]$ and changes success by −4.4 pp, mostly with clutter (interaction $[+0.4, +7.0]$).

Search versus its base policy. Removing search changes L by $+0.6$ $[-1.3, +2.5]$ overall, but its action interaction is resolved ($[+0.6, +7.3]$): removing search raises L with 26 commands ($+2.6$), while the change with 6 commands is unresolved (-1.4). In the regression and generated suites, search rescues 5 multi-drone episodes and loses 13 and removing it changes L by -2.4 $[-3.7, -0.9]$ (Appendix A.7). With value guidance, search has no consistent completion benefit across the evaluated suites; the measured performance is largely explained by the guided base policy.

S4: search budget

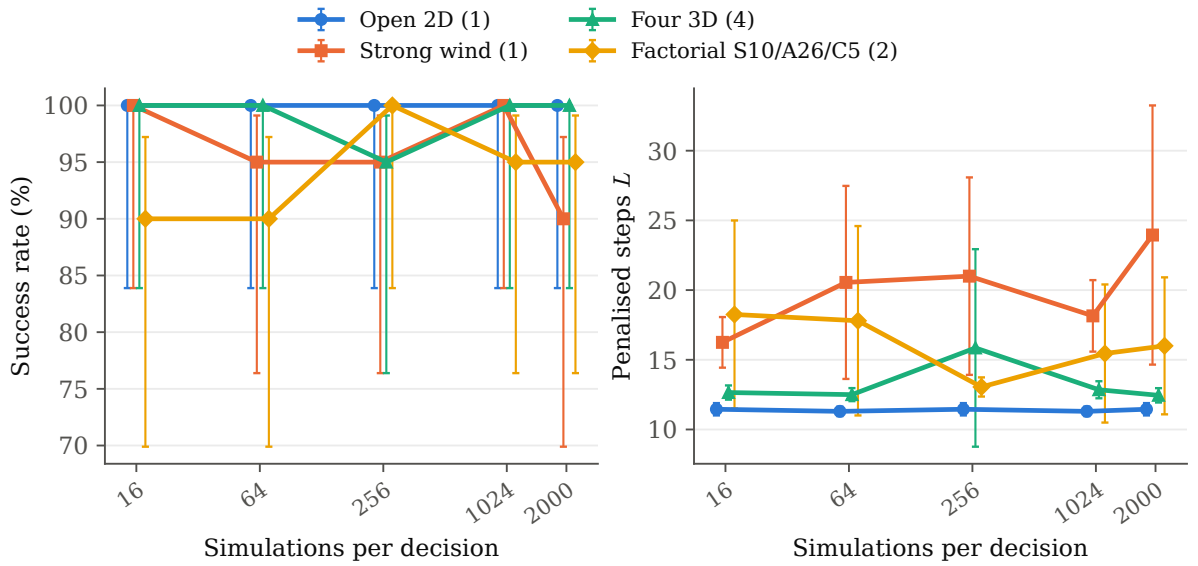


Figure 4: Success and L versus simulations per decision; 20 runs per point, with Wilson and t 95% intervals. Drone counts are in parentheses.

There is no consistent improvement from 16 to 2,000 simulations. For four drones, success is 20/20 and 20/20 respectively, while mean new-node depth increases from 2.1 to 3.0. More search changes the tree without a demonstrated completion benefit here; overlapping intervals do not establish budget equivalence. Appendix A.6 also reports admitted root actions and new-node depth, distinguishing tree growth from task performance.

Dijkstra versus value guidance. Replacing both Dijkstra guidance and its distance-based leaf estimate with VI gives $1540/1,680$ successes versus $1539/1,680$, with paired $\Delta L = -0.8$ $[-1.4, -0.3]$ steps. The success-change interval includes zero; this comparison does not isolate guidance from leaf estimation (Appendix A.8).

S5: generated layouts

Across the crossed drone-count/noise set, 144/160 episodes succeed. At $\alpha \leq 0.25$, 78/80 runs succeed, including 20/20 with four drones; four drones at $\alpha = 2$ succeed in 6/10. This indicates sensitivity to noise in the tested layouts, while increasing N jointly enlarges state and action spaces. Mean new-node depth is 3.1 for one drone and 2.5 for four, consistent with less successor reuse when joint outcomes multiply. Each (N, α) pair has only one layout, so the per-layout results in Appendix A.10 do not isolate layout effects. For the four-drone layout at $\alpha = 2$, joint value iteration was not attempted ($140^4 \approx 4 \times 10^8$ joint positions per step). Each drone alone could achieve at least 94% success when other drones are removed. Executing those policies together gives 16% success (1,000 diagnostic runs; Wilson 95% interval [13.8, 18.3]%), compared

with 6/10 for the solver. The different sample sizes and unpaired comparison do not establish a solver advantage or isolate the cause of failure. They motivate further coordination tests (`results/ceilings.json`).

Limitations

The factorial covers two drones at $\alpha = 0.25$, and S5 has only one layout per (N, α) pair. State padding changes available space, and richer commands change control authority; the ranking is specific to these levels. The t intervals for bimodal L are approximate; all intervals are pointwise, without correction for multiple comparisons. Fixed-simulation runs share the demo's trial cap, but omit its binding time limit (73 of 74 local demo decisions reached the trial cap). Slower hardware could cut simulations; S4 does not establish budget equivalence. Single-drone values ignore the other drones: the leaf checks disconnection by frozen drones but not detours around them, and inter-drone risk enters only through the proposal surrogate. Successful wall-gap runs average 20 steps for an 11-step static route, against 39 with the risk-free cutoff estimate of guidance A. This is consistent with value cutoffs removing hesitation before an unavoidable risky passage (Figure 1, bottom right), although these experiments do not isolate the cause. Finite-horizon value iteration on the supplied model gives maximum success probabilities of 70.5% for the wall gap and 63.1% for the corridor (Appendix A.9). Both lie inside the Wilson intervals of the solver's 14/20 and 13/20 successes. This demonstrates that some slip risk is unavoidable, but does not establish optimality or attribute every observed failure to it. Outcome progressive widening or an open-loop tree are possible follow-up designs for improving successor reuse; neither was evaluated here.

A Supporting definitions and results

Complete regression, budget and tuning data remain in the source archive. All evaluation suites were run with the submitted planner, whose hash is recorded in each `results/*.meta.json`; the earlier Dijkstra-guidance runs are archived in `results/guidance_dijkstra/` (Appendix A.8).

A.1 Algorithm definitions

For a free in-grid target command, let $\mathcal{N}(q)$ be the physical neighbours of q and $V(q) = |\mathcal{N}(q)|$. The supplied motion law gives

$$\pi(q) = \frac{d(q)}{d(q)+\alpha}, \quad M(q) = \pi(q) + (V(q)-1)\frac{1-\pi(q)}{B-1},$$
$$\ell(q) = \frac{\pi(q)}{M(q)}, \qquad \sigma(q) = \frac{1-\pi(q)}{(B-1)M(q)}.$$

Value guidance (submitted). For command a from free cell p, the intended neighbour has weight $\pi(p)$, every other in-grid neighbour $(1-\pi(p))/(B-1)$, the row is renormalised, and mass on obstacle cells is a terminal crash. Value iteration uses these exact rows (stored sparsely) with $r_i = c_{\text{step}} + C\Pr(\text{crash}) + G\Pr(\text{arrival at } g_i)$ and $V_{i,h}(g_i) = 0$, because arrival is terminal for that drone. Per-horizon action values are kept when $nA(H+1)N \le 2.5 \times 10^7$ (n free cells, A commands) on grids of at most 20,000 cells; otherwise the planner uses the Dijkstra guidance below. On the demo maps this takes under 0.2 s before the first decision.

Dijkstra guidance (comparison A). Entry costs are $1 + \lambda\rho(q)$ for a non-goal cell and 1 for

the goal, where

$$\rho(q) = |\mathcal{N}(q) \cap O|\sigma(q), \qquad \lambda = \frac{w(G + |C|)}{\max\{|c_{\text{step}}|, 10^{-3}(G + |C|), 10^{-9}\}}, \quad w = 1.$$

With $w = 0$, these are breadth-first distances. The supplied empty-map distance transform uses a virtual point beyond the grid corner, so slips remain possible without obstacles.

Grid boundaries are not obstacles: out-of-grid outcome mass is removed before normalisation. The simulator encodes a joint command as

$$a_{\text{int}} = \sum_{i=1}^{N} a_i B^{i-1}.$$

Maps are built for at most 150,000 cells. Larger grids use Chebyshev distance without a risk map; fixed-altitude goals on another layer remain unreachable. Frozen-goal disconnection checks are cached by drone and reached subset for at most 20,000 cells. Above that size, the leaf uses static reachability. These fallbacks are implementation limits, not guarantees of safe coordination.

A rollout cut off after L sampled steps returns

$$\sum_{j=0}^{L-1} \gamma^j r_j/\kappa + \gamma^L \widehat{V}(s_L, h - L)/\kappa,$$

with zero continuation on termination. For an edge sample v, update

$$n \leftarrow n + 1, \quad \Delta \leftarrow v - Q, \quad Q \leftarrow Q + \Delta/n, \quad S \leftarrow S + \Delta(v - Q).$$

The pooled root variance is $\hat{\sigma}^2 = \sum_a S_a / \sum_a (n_a - 1)$ over visited actions, or zero when the denominator is zero.

A.2 Development-set parameter selection

Sixteen layouts cross $N = 1, 2, 3, 4$ with $\alpha \in \{0.25, 0.5, 1, 2\}$ on $10 \times 10 \times 1$ or $10 \times 10 \times 5$ grids, with 0–5 obstacles (generator seed 4620). Each setting uses ten execution seeds and at most 500 simulations per decision; search-free settings use none. These 2,880 development episodes are separate from all 3,460 evaluation episodes.

Round 1 compares risk weights, UCT constants and search-free policies under most-visited root choice and widening coefficient 1.5. Round 2 compares the anchored rule, its threshold and rollout exploration, and widening coefficient 0.5. The selected setting is `anchor_z1_w0.5`: risk weight 1, UCT constant 0.35, switch threshold 1, widening coefficient 0.5 and rollout exploration 0.1. Here the suffix `w0.5` denotes widening, not the risk weight. Round 3 compares that Dijkstra guidance with single-drone value-iteration guidance, each with the anchored rule and with largest-mean-Q root choice, plus both search-free policies. The selection criterion required no worse observed success count or mean L. Value guidance met it (143/160 versus 140/160 successes), whereas largest-mean-Q choice raised L under both guidances (130/160 and 137/160). The round-3 Dijkstra arm reproduces the round-2 selected setting exactly. Development tables retain the planner hashes recorded for those versions; all main evaluation suites were rerun with the final implementation. Exact sources for every recorded planner hash are indexed in `provenance/planner_sources.json`.

Table 3: Development results: 160 episodes per setting. L: mean ±95% interval half-width. ΔL: paired change from the selected setting, with a 95% interval over ten seed blocks; lower is better. Selection across 18 settings makes these exploratory comparisons optimistic for the development layouts.

Round	Setting	Success	L	ΔL (95% CI)
1	risk0	125/160	24.3 ± 3.1	+5.3 [+2.7, +8.0]
1	risk0.3	130/160	23.4 ± 2.9	+4.5 [+2.2, +6.7]
1	risk1	129/160	23.4 ± 3.0	+4.5 [+1.4, +7.6]
1	risk3	126/160	23.8 ± 3.0	+4.9 [+1.7, +8.0]
1	risk1_c0.1	129/160	23.0 ± 2.9	+4.1 [+2.3, +6.0]
1	risk1_c1	123/160	25.1 ± 3.2	+6.2 [+4.3, +8.1]
1	greedy_risk0	121/160	23.9 ± 3.2	+5.0 [+2.7, +7.3]
1	greedy_risk1	138/160	19.8 ± 2.6	+0.9 [-1.2, +2.9]
2	anchor_z1	133/160	21.3 ± 2.8	+2.4 [+1.1, +3.7]
2	anchor_z2	136/160	20.7 ± 2.7	+1.8 [-0.2, +3.7]
2	anchor_z1_eps0	133/160	21.4 ± 2.8	+2.5 [+1.3, +3.7]
2	anchor_z1_w0.5	140/160	19.7 ± 2.5	+0.8 [-0.4, +2.1]
3	dijkstra	140/160	19.7 ± 2.5	+0.8 [-0.4, +2.1]
3	vi	143/160	18.9 ± 2.4	—
3	dijkstra_maxq	130/160	23.3 ± 3.0	+4.4 [+1.7, +7.2]
3	vi_maxq	137/160	21.8 ± 2.8	+2.9 [+1.5, +4.4]
3	greedy_dijkstra	138/160	19.8 ± 2.6	+0.9 [-1.2, +2.9]
3	greedy_vi	142/160	18.3 ± 2.4	-0.6 [-2.1, +0.8]

A.3 Factorial results

Table 4: All 27 S1 conditions, $n = 20$ each. L and Return: mean ±95% interval half-width; Calls: mean simulator calls per decision.

n_y	$\lvert A\rvert$	$\lvert O\rvert$	Success	L	Return	Calls
5	36	0	20/20	13.9 ± 0.7	132.4 ± 2.6	17,969
5	36	2	20/20	16.9 ± 1.1	119.5 ± 3.8	18,171
5	36	5	20/20	18.9 ± 1.3	112.0 ± 4.3	18,294
5	324	0	20/20	11.9 ± 0.5	140.4 ± 1.8	14,984
5	324	2	20/20	13.2 ± 0.8	136.0 ± 2.8	15,991
5	324	5	19/20	15.8 ± 4.9	122.5 ± 24.9	15,948
5	676	0	20/20	11.8 ± 0.5	140.8 ± 1.6	14,693
5	676	2	17/20	20.1 ± 8.1	100.6 ± 40.9	16,180
5	676	5	20/20	13.6 ± 1.0	134.4 ± 3.4	15,345
10	36	0	20/20	14.6 ± 0.6	129.9 ± 2.4	17,897
10	36	2	19/20	19.8 ± 4.6	106.2 ± 23.3	18,345
10	36	5	20/20	19.4 ± 0.9	109.8 ± 3.4	18,786
10	324	0	20/20	11.8 ± 0.5	140.8 ± 1.7	14,980
10	324	2	20/20	12.7 ± 0.6	137.7 ± 2.1	15,687
10	324	5	20/20	13.2 ± 0.7	135.4 ± 2.7	15,871
10	676	0	20/20	11.8 ± 0.6	140.7 ± 1.8	14,763
10	676	2	19/20	14.9 ± 5.0	121.4 ± 35.5	15,605
10	676	5	19/20	16.0 ± 4.9	121.8 ± 24.8	16,112
20	36	0	20/20	14.5 ± 0.7	130.3 ± 2.5	17,909

Continued on next page

Table 4 continued

n_y	$\lvert A\rvert$	$\lvert O\rvert$	Success	L	Return	Calls
20	36	2	18/20	21.8 ± 6.2	95.7 ± 31.9	18,429
20	36	5	19/20	22.6 ± 4.4	95.9 ± 22.3	19,038
20	324	0	20/20	11.8 ± 0.5	140.7 ± 1.7	14,980
20	324	2	19/20	15.0 ± 5.0	126.0 ± 25.2	16,044
20	324	5	19/20	15.3 ± 5.0	124.0 ± 25.0	15,930
20	676	0	20/20	11.8 ± 0.5	140.7 ± 1.6	14,725
20	676	2	19/20	14.8 ± 5.0	126.8 ± 25.3	15,518
20	676	5	20/20	13.1 ± 0.7	135.9 ± 2.7	15,639

A.4 Mechanism controls

Command masks restrict intended directions; the physical slip law retains all 26 neighbours. Duplicated labels leave the physical commands unchanged.

Table 5: S2 clutter controls ($n_y = 10$, $n = 20$). L: mean ±95% interval half-width; Obst./Veh.: collision episodes.

Commands	α	$\lvert O\rvert$	Success	L	Obst.	Veh.
all 26	0.25	0	20/20	11.8 ± 0.6	0	0
		2	19/20	14.9 ± 5.0	0	1
		5	19/20	16.0 ± 4.9	1	0
	0.01	0	20/20	11.1 ± 0.1	0	0
		2	19/20	13.6 ± 5.1	1	0
		5	20/20	11.2 ± 0.3	0	0
axial 6	0.25	0	20/20	14.6 ± 0.6	0	0
		2	19/20	19.8 ± 4.6	1	0
		5	20/20	19.4 ± 0.9	0	0
	0.01	0	20/20	12.9 ± 0.4	0	0
		2	20/20	13.3 ± 0.3	0	0
		5	20/20	14.7 ± 0.5	0	0

Table 6: Duplicated labels ($n_y = 10$, $\lvert O\rvert = 2$, $n = 20$). Root: admitted actions; Selected: visit share of the executed action. L: mean ±95% interval half-width.

Labels	$\lvert A\rvert$	Success	L	Root	Selected (%)
×1	676	19/20	14.9 ± 5.0	22.3	5.7
×2	2,704	19/20	14.7 ± 5.0	23.0	5.1
×4	10,816	19/20	14.6 ± 5.0	23.0	5.0

A.5 Component effects

Table 7: Overall paired S3 effects, with 95% intervals. Uniform proposals: ten seeds; others: twenty.

Variant	ΔL	Δsuccess (pp)
Uniform proposals/rollouts	+35.7 [+33.6, +37.8]	-50.0 [-54.2, -45.8]
Zero leaf value	+0.5 [-1.0, +2.0]	-1.9 [-5.3, +1.6]
No progressive widening	+1.0 [-0.5, +2.5]	-2.5 [-5.6, +0.6]
No inter-drone risk term	+1.8 [+0.1, +3.5]	-4.4 [-8.3, -0.5]
Most-visited final action	+3.2 [+2.1, +4.3]	-1.2 [-3.9, +1.4]
Largest mean-Q final action	+3.7 [+2.4, +5.1]	-3.8 [-7.6, +0.1]
Guided policy, no search	+0.6 [-1.3, +2.5]	-3.8 [-8.0, +0.5]

Table 8: Mean paired ΔL at low/high complexity levels. *: the interaction's 95% interval excludes zero.

Variant	n_y: 5 / 20	$\lvert A\rvert$: 36 / 676	$\lvert O\rvert$: 0 / 5
Uniform proposals/rollouts	+37.7 / +33.8	+34.0 / +37.5	+29.1 / +42.3*
Zero leaf value	+0.7 / +0.3	+0.1 / +0.9	-0.3 / +1.3
No progressive widening	+1.1 / +0.9	-0.3 / +2.4	-0.1 / +2.1
No inter-drone risk term	+2.1 / +1.5	+1.3 / +2.3	-0.1 / +3.7*
Most-visited final action	+3.8 / +2.6	+5.7 / +0.7*	+0.7 / +5.7*
Largest mean-Q final action	+4.8 / +2.6	+6.5 / +1.0*	+0.7 / +6.8*
Guided policy, no search	+0.6 / +0.6	-1.4 / +2.6*	-0.5 / +1.7

Table 9: S3 successes at all eight factorial corners. Uniform proposals: $n = 10$ per corner; all other rows: $n = 20$. Within each n_y group, columns give intended commands per drone / obstacle count. These counts complement the paired effects and interaction intervals above.

	$n_y = 5$				$n_y = 20$			
Variant	6/0	6/5	26/0	26/5	6/0	6/5	26/0	26/5
Full solver	20/20	20/20	20/20	20/20	20/20	19/20	20/20	20/20
Uniform proposals/rollouts	10/10	0/10	10/10	0/10	10/10	0/10	9/10	0/10
Zero leaf value	20/20	20/20	20/20	18/20	20/20	18/20	20/20	20/20
No progressive widening	20/20	19/20	20/20	19/20	20/20	20/20	20/20	17/20
No inter-drone risk term	20/20	18/20	20/20	18/20	20/20	18/20	20/20	18/20
Most-visited final action	20/20	19/20	20/20	19/20	20/20	19/20	20/20	20/20
Largest mean-Q final action	20/20	16/20	20/20	19/20	20/20	18/20	20/20	20/20
Guided policy, no search	20/20	19/20	20/20	18/20	20/20	19/20	20/20	17/20

A.6 Search-budget diagnostics

Root branching and mean new-node depth measure how the search changes as the simulation cap grows. New-node depth is a sampled mean, not the maximum configured tree depth of eight. The four-drone tree remains shallow even at 2,000 trials, consistent with limited reuse of stochastic joint successors.

Table 10: S4 diagnostics, $n = 20$ per row. L: mean ±95% interval half-width; Root: mean admitted actions; Depth: mean new-node depth. Drone counts are in parentheses.

Task (drones)	Sims	Success	L	Root	Depth
Open 2D (1)	16	20/20	11.4 ± 0.4	2.0	2.49
Open 2D (1)	64	20/20	11.3 ± 0.3	4.0	3.09
Open 2D (1)	256	20/20	11.4 ± 0.4	7.7	3.54
Open 2D (1)	1,024	20/20	11.3 ± 0.3	7.7	4.12
Open 2D (1)	2,000	20/20	11.4 ± 0.4	7.7	4.43
Strong wind (1)	16	20/20	16.2 ± 1.8	2.0	1.65
Strong wind (1)	64	19/20	20.6 ± 6.9	4.0	1.84
Strong wind (1)	256	19/20	21.0 ± 7.1	8.0	2.03
Strong wind (1)	1,024	20/20	18.1 ± 2.6	13.9	2.27
Strong wind (1)	2,000	18/20	23.9 ± 9.3	15.1	2.45
Four 3D (4)	16	20/20	12.7 ± 0.5	2.0	2.13
Four 3D (4)	64	20/20	12.5 ± 0.5	4.0	2.48
Four 3D (4)	256	19/20	15.8 ± 7.1	8.0	2.70
Four 3D (4)	1,024	20/20	12.8 ± 0.6	15.8	2.90
Four 3D (4)	2,000	20/20	12.4 ± 0.5	22.3	2.96
Factorial S10/A26/C5 (2)	16	18/20	18.2 ± 6.7	2.0	2.19
Factorial S10/A26/C5 (2)	64	18/20	17.8 ± 6.8	4.0	2.59
Factorial S10/A26/C5 (2)	256	20/20	13.1 ± 0.7	8.0	2.82
Factorial S10/A26/C5 (2)	1,024	19/20	15.4 ± 5.0	15.9	3.08
Factorial S10/A26/C5 (2)	2,000	19/20	16.0 ± 4.9	22.4	3.19

A.7 Regression coverage and paired baseline

The 18 named scenarios each use 20 seeds and a cap of 2,000 simulations. They cover initial goal occupancy, swaps, boundaries, small or thin grids, unreachable goals, a short horizon, narrow passages, altitude changes, strong noise and multi-drone motion. The three statically infeasible cases and the one-step horizon are expected failures; collisions on feasible tasks remain recorded failures.

Table 11: Named regression scenarios. No search: paired base policy; Obst./Veh.: collision episodes. L: mean ±95% interval half-width, using each scenario's own horizon as the failure penalty. Calls/step counts all generative-model calls, including rollouts.

Scenario	Success	No search	Obst.	Veh.	L	Calls/step
`demo_open_2d`	20/20	20/20	0	0	11.4 ± 0.4	16,403
`demo_four_3d`	20/20	20/20	0	0	12.4 ± 0.5	14,955
`already_at_goals`	20/20	20/20	0	0	0.0 ± 0.0	0
`single_cell_goal`	20/20	20/20	0	0	0.0 ± 0.0	0
`one_reached`	20/20	20/20	0	0	8.3 ± 0.5	14,237
`swap_2d`	20/20	20/20	0	0	5.2 ± 0.3	10,328
`boundary_reverse`	20/20	20/20	0	0	5.3 ± 0.3	11,202
`tiny_2d`	20/20	20/20	0	0	1.0 ± 0.0	3,872
`thin_1d`	20/20	20/20	0	0	7.0 ± 0.0	12,733
`infeasible_altitude`	0/20	0/20	0	0	80.0 ± 0.0	0
`infeasible_wall`	0/20	0/20	0	0	80.0 ± 0.0	0
`infeasible_shared_goal`	0/20	0/20	0	0	80.0 ± 0.0	0

Continued on next page

Table 11 continued

Scenario	Success	No search	Obst.	Veh.	L	Calls/step
`horizon_one`	0/20	0/20	0	0	1.0 ± 0.0	2,000
`narrow_passage`	15/20	17/20	5	1	30.6 ± 13.7	17,683
`altitude_detour`	20/20	20/20	0	0	9.2 ± 0.3	14,274
`strong_wind`	18/20	19/20	2	0	23.9 ± 9.3	16,796
`three_3d`	20/20	20/20	0	0	10.6 ± 0.5	13,770
`four_2d`	20/20	20/20	0	0	12.6 ± 0.8	15,334

Among 240 feasible multi-drone regression/generated pairs, search succeeds in 221 versus 229; removing search changes L by -2.4 $[-3.7, -0.9]$. Search rescues 5 episodes and loses 13. On the 180 single-drone pairs, search succeeds in 156 versus 158 ($\Delta L = -0.9$ $[-1.8, -0.2]$). These exploratory intervals resample whole execution-seed clusters, retaining the episode-weighted means across layouts; they are distinct from the balanced S1–S3 seed-block t intervals. Removing search changes multi-drone success by $+3.3$ pp $[+0.4, +6.1]$. Both the multi-drone success and completion-length intervals favour the base policy on these fixed layouts, without establishing a general ordering. These comparisons exclude 100 zero-step episodes (already at the goals or rejected as statically infeasible) from the 520 baseline pairs.

An exhaustive small-model check separately verifies 704 state-action pairs and 6,048 positive-probability outcomes on all 11 admissible nonterminal states of a free $2 \times 2 \times 1$ two-drone grid (`results/small_model_verification.json`). This checks model semantics on that finite example, not performance on arbitrary maps.

A.8 Value-iteration versus Dijkstra guidance

S1, S2, S4, S5 and the regression/supplementary suites compare the earlier Dijkstra guidance and distance-based leaf estimate (A) with submitted VI guidance and leaf values (B) on the same scenarios and seeds; pairs match scenario, variant, simulation cap and seed. S3 is omitted here because removing risk terms has different meanings under A and B. The A runs are archived in `results/guidance_dijkstra/`. Across the full-solver suites, B succeeds in 1540 of $1,680$ distinct paired episodes and A in 1539: B rescues 44 and loses 43. The success change is $+0.1$ pp $[-0.9, +1.1]$; penalised length changes by -0.8 $[-1.4, -0.3]$ steps. The aggregate excludes the separate 520-pair search-free baseline row and 80 repeated conditions shared by the budget and other suites. Intervals use 20,000 execution-seed cluster bootstrap resamples, rather than treating all episode differences as independent. Value guidance was selected on the development set before these evaluation runs; the comparison does not establish uniform superiority over Dijkstra guidance. The pooled completion-length interval favours value guidance, while the success-change interval includes zero.

Table 12: Paired comparison of guidance and leaf values B (VI, submitted) with A (Dijkstra and a distance-based leaf estimate). Rescues/Losses: pairs that only B / only A completes; ΔL: B minus A with a 95% seed-cluster bootstrap interval. The full-solver aggregate excludes the search-free baseline row and counts repeated conditions once.

Suite	Pairs	A	B	Rescues	Losses	ΔL (95% CI)
S1 factorial	540	527	527	12	12	-0.88 [-1.63, -0.04]
S2 controls	160	160	157	0	3	+0.94 [+0.03, +2.35]
S4 budget	400	381	389	17	9	-2.08 [-3.55, -0.72]
S5 layouts	160	148	144	8	12	+1.12 [-0.53, +2.74]
Regression	360	275	273	3	5	+0.19 [-0.66, +1.05]
Supplementary	140	125	127	6	4	-3.77 [-6.04, -1.41]
Base policy, no search	520	415	427	17	5	-1.38 [-2.22, -0.50]
Combined (unique)	1680	1539	1540	44	43	-0.82 [-1.38, -0.28]

A.9 Supplementary end-to-end cases

The supplementary cases below include fixed-altitude single-drone maps, separate altitude layers and a goal inside a teammate's corridor. Wall-gap and corridor cases use seven/eight obstacles, beyond the demo's five-obstacle range. No timeout occurred; all failures were collisions. The DP maximum is the largest success probability any policy can achieve within H under the supplied model, computed by finite-horizon value iteration (`scripts/compute_ceilings.py`; the two-layer crossing case is a product because drones on different fixed layers never interact; the four-drone case was not solved). This optimises success probability, whereas B1 optimises discounted reward. The computed maxima lie inside the solver's Wilson intervals, which does not establish policy equivalence.

Table 13: Supplementary cases ($n = 20$ per policy). 95% CI: Wilson interval of the success rate; No search: paired base policy; Obst./Veh./T.o.: full-solver failure counts. L: mean ±95% interval half-width.

Case	Success	95% CI (%)	No search	Obst.	Veh.	T.o.	L
Supplied map, 1 drone	20/20	[84, 100]	20/20	0	0	0	10.7 ± 0.9
Wall with one gap, 1 drone	14/20	[48, 85]	14/20	6	0	0	37.7 ± 13.5
Slalom, 1 drone	20/20	[84, 100]	19/20	0	0	0	20.1 ± 1.6
Five obstacles, 1 drone	20/20	[84, 100]	20/20	0	0	0	16.4 ± 1.8
Two layers, crossing, 2 drones	20/20	[84, 100]	20/20	0	0	0	12.8 ± 1.3
Two layers, 4 drones	20/20	[84, 100]	19/20	0	0	0	13.4 ± 0.7
Goal inside corridor, 2 drones	13/20	[43, 82]	13/20	7	0	0	35.1 ± 15.8

Table 14: Enumerated finite-horizon success maxima (floating-point arithmetic) and policy validation in the supplied simulator. Five policies use seeds 0–1,999; the corridor adds a fixed batch of 8,000 disjoint seeds, giving 10,000 runs in total. The initial corridor interval missed its DP value; the original batch is retained in `results/ceilings.json` and the extension in `results/ceiling_validation.json`. Wilson 95% intervals describe these diagnostic samples, separate from the 3,460 solver-evaluation episodes; they do not establish policy equivalence.

Case	DP max. (%)	Policy successes	95% CI (%)
Supplied map, 1 drone	99.9	1,998/2,000	[99.6, 100.0]
Wall with one gap, 1 drone	70.5	1,382/2,000	[67.0, 71.1]
Slalom, 1 drone	88.0	1,754/2,000	[86.2, 89.1]
Five obstacles, 1 drone	95.8	1,934/2,000	[95.8, 97.4]
Two layers, crossing, 2 drones	99.9	1,997/2,000	[99.6, 99.9]
Goal inside corridor, 2 drones	63.1	6,300/10,000	[62.0, 63.9]

A.10 Generated-layout results

Table 15: All 16 S5 layouts, one per (N, α) pair, ten seeds each. L: mean ±95% interval half-width; Depth: mean new-node depth.

N	α	Success	Wilson 95% (%)	L	Depth
1	0.05	10/10	[72, 100]	7.0 ± 0.0	3.84
1	0.25	10/10	[72, 100]	7.4 ± 0.6	3.52
1	1	10/10	[72, 100]	9.2 ± 1.5	2.79
1	2	8/10	[49, 94]	19.4 ± 15.4	2.45
2	0.05	10/10	[72, 100]	7.7 ± 0.7	3.55
2	0.25	10/10	[72, 100]	8.4 ± 1.3	3.15
2	1	10/10	[72, 100]	11.8 ± 2.2	2.40
2	2	6/10	[31, 83]	34.5 ± 15.9	2.00
3	0.05	10/10	[72, 100]	8.3 ± 0.8	3.34
3	0.25	8/10	[49, 94]	19.7 ± 15.2	2.76
3	1	10/10	[72, 100]	11.3 ± 1.5	2.17
3	2	10/10	[72, 100]	17.7 ± 3.2	1.88
4	0.05	10/10	[72, 100]	7.8 ± 0.8	3.32
4	0.25	10/10	[72, 100]	8.5 ± 0.8	2.99
4	1	6/10	[31, 83]	31.6 ± 17.5	1.88
4	2	6/10	[31, 83]	36.8 ± 14.5	1.74

B Example trajectories and planning times

Figure 5 spans four pages and shows stored seed-0 full-solver traces, including a collision; these examples do not replace B4's repeated-run statistics. Squares mark starts, stars goals, and crosses unfinished drones' returned positions. Collision targets are not logged. Each row identifies either the demo's 9.5 s allowance or a fixed-simulation experiment with a relaxed deadline. The wall-gap and corridor layouts have seven and eight obstacles, beyond the demo's five-obstacle range. The complete gallery covers all 76 evaluation configurations and the supplied example, excluding development layouts.

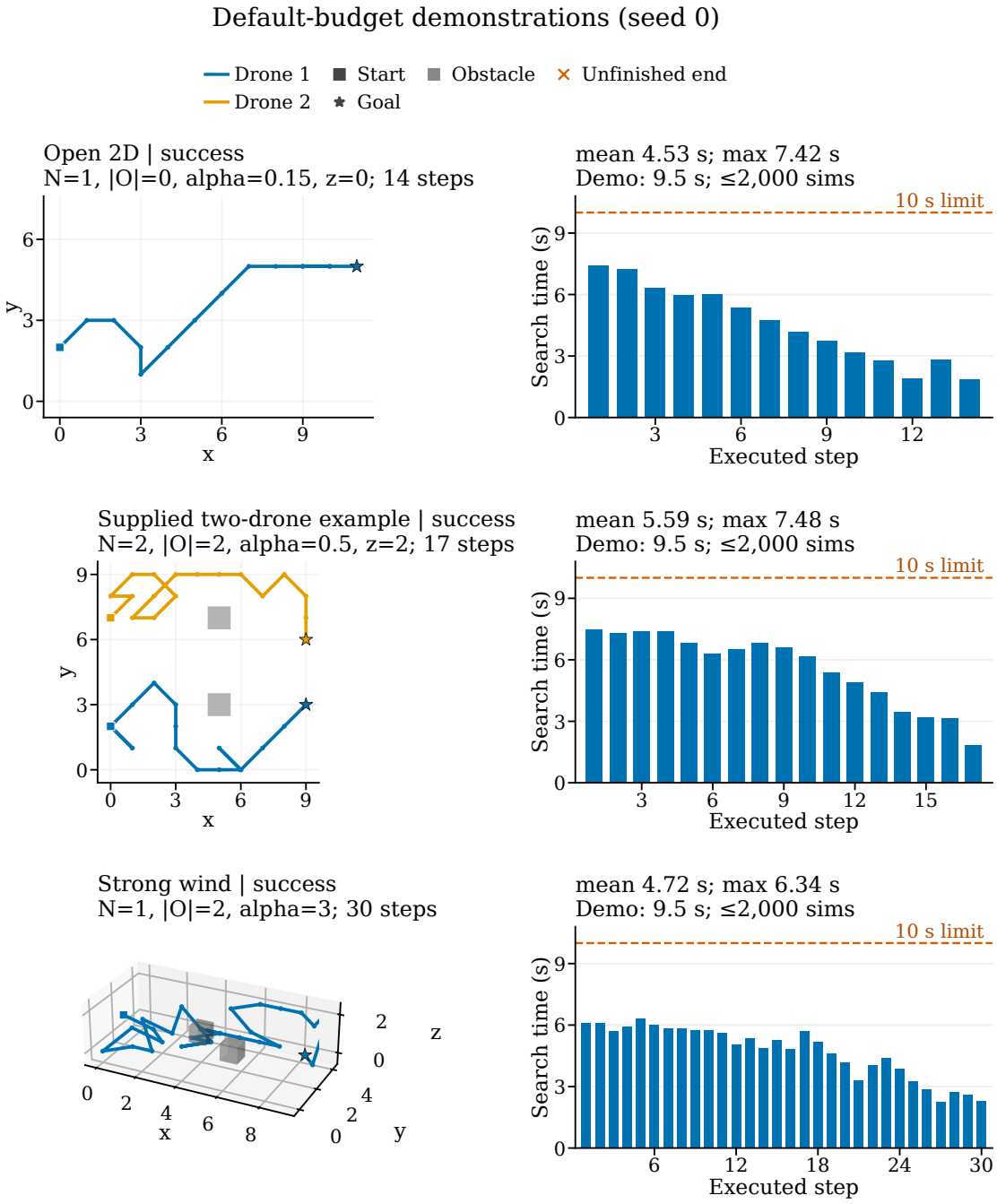


Figure 5: Representative trajectories (left) and per-step planning times (right) for twelve scenarios.

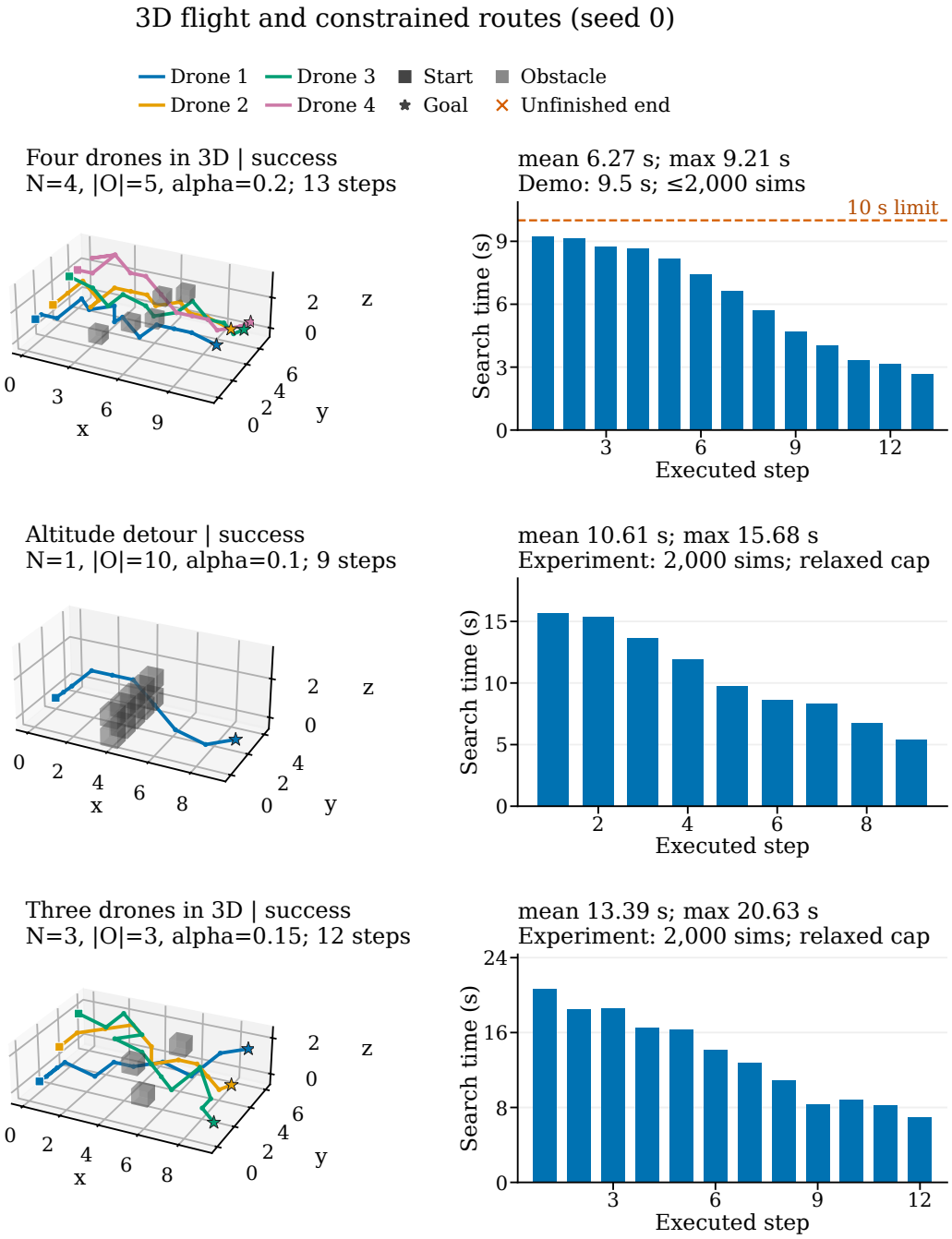


Figure 5: *Continued.*

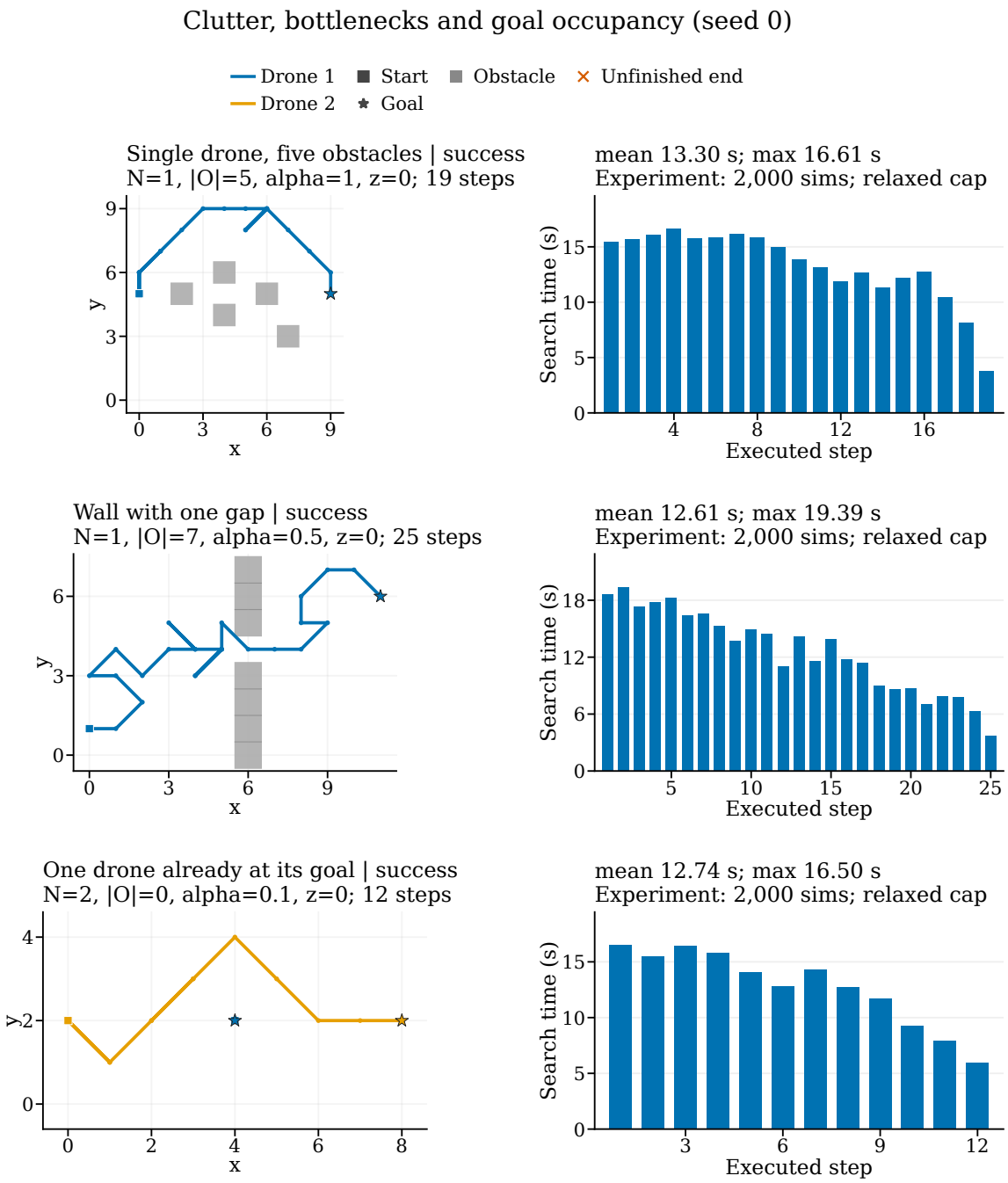


Figure 5: *Continued.*

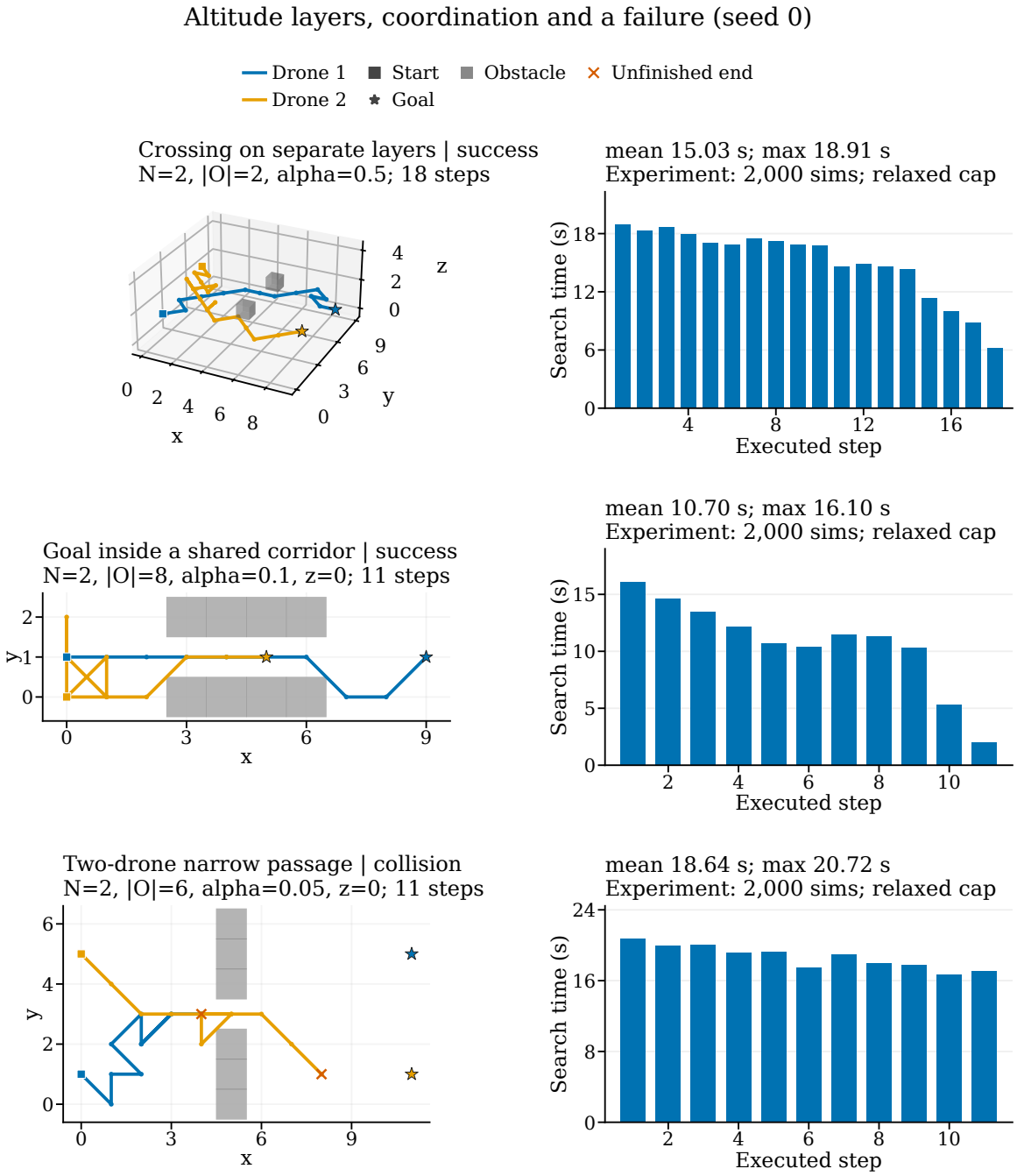


Figure 5: *Continued.*